Kaicheng Xue

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RESEARCH INTERESTS

Understanding and evaluating the behavior of large language models, from their internal mechanisms to their use as agents in the real world.

EDUCATION

University of Virginia, Charlottesville, VA Aug 2025 – Dec 2026 (expected)
M.S. in Computer Science, GPA: 4.0/4.0
Thesis: *Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy*
Advisor: Yangfeng Ji
Relevant Coursework: Machine Learning, Natural Language Processing, Trustworthy AI, Probabilistic Machine Learning, Data Privacy, Neural Networks, Geometry of Data

University of Wisconsin–Madison, Madison, WI Sept 2021 – May 2025
B.S. in Mathematics and Computer Science (double major), GPA: 3.79/4.0 (Mathematics 3.81, Computer Science 3.78)
Relevant Coursework: Introduction to Algorithms, Introduction to Artificial Intelligence, Real Analysis, Stochastic Processes, Probability, Abstract Algebra, Linear Optimization, Linear Algebra, Ordinary Differential Equations, Discrete Mathematics
Honors: College of Letters & Science Dean's List

PUBLICATIONS

Xue, K., Zhang, X., Mao, R., & Cambria, E. "Understanding the Hidden State of Language Models from the Perspective of Sentiment Analysis." In *Proceedings of the 2024 IEEE International Conference on Data Mining Workshops (ICDMW)*, 2024. *(Workshop paper)*

RESEARCH EXPERIENCE

Graduate Research, University of Virginia Jan 2026 – Nov 2026
Evaluating Entropy and Spectral Gap of Hidden-State Markov Chains as Predictors of LLM Reasoning Accuracy (Advisor: Yangfeng Ji)

- Tested whether hidden-state dynamics over chain-of-thought steps explain reasoning accuracy beyond model size and family: modeled layer-wise trajectories as discrete Markov chains, derived per-layer entropy rates from stationary distributions, and characterized mixing via spectral gaps of transition matrices
- Evaluated nine Llama and Qwen3 models, plus LoRA-fine-tuned variants of Llama-3.2-3B, across four reasoning benchmarks (HotpotQA, ProofWriter, GSM8K, HSH-Math)
- Designed a rigorous evaluation protocol using nested leave-one-out cross-validation with in-fold feature selection and selection-aware permutation tests
- Found that model size and family explain most of the variation in accuracy across models on two of four datasets, but relationships learned on non-fine-tuned models register the large accuracy gains of LoRA-fine-tuned variants of a fixed base model (a preliminary result on five fine-tuned models); the discovered states mostly reflect a step's position in the chain, motivating content-based states and answer-level prediction as next steps

Student Researcher, University of Virginia Aug 2026 – Present
LLM Abuse in Digital Advertising Marketplaces: Detection and Mitigation
PI: Tianhao Wang (CS); Co-PIs: Michael Albert (Darden), Jingjing Li (McIntire)

- Build an end-to-end simulation of an Amazon-style marketplace in which $\sim$200 LLM shopper agents per target browse, compare, and purchase products, to measure how coordinated fake-review attacks shift purchasing behavior at the population level
- Ground the environment in real data (matched personas to real Amazon buyers, demographics calibrated to U.S. BLS survey data) and implement a coordinated LLM-generated fake-review attack that combines push, segment-targeted, and average-profile imitation
- Run baseline-vs.-attack experiments across 40 targets and cross-model comparisons across 8 LLMs; preliminary results show that attack effects concentrate on targets reaching the first results page and that agents cite fake reviews without expressing suspicion

Course Research Project, University of Virginia Spring 2026

Privacy Attack via Temporal Embedding Inversion on RAG Systems

- Proposed a temporal-aware sliding-window method for inverting conversation embeddings leaked from RAG systems, incorporating dialogue history to improve text reconstruction and sensitive-information extraction; constructed a 3,000-conversation dataset with three categories of embedded private information (user profile, financial, medical) to evaluate attack success
- Analyzed why a non-linear extension (Temporal Context Fusion Network) fails to improve over the linear sliding-window approach, identifying that non-linear projection distorts embedding-space geometry rather than improving alignment
- Achieved best attack success rates of 41% (user profile) and 53% (medical) with the fine-tuned sliding-window variant

Research Intern, Nanyang Technological University May 2024 – Sept 2024

Internal Mechanisms of LLMs in Sentiment Analysis (Supervisors: Xulang Zhang, Rui Mao, Erik Cambria)

- Analyzed hidden states of a fine-tuned RoBERTa model at all 12 layers and at the 12 attention heads of layer 11, using K-means, spectral, and hierarchical clustering to probe sentiment encoding
- Found that clustering separates sentiment well mainly in the last three layers ($\sim$80% accuracy at layer 12 versus $\sim$50–54% in most earlier layers), and that 3 of the 12 heads in layer 11 reach only $\sim$50–54% versus $\sim$78–79% for most others, indicating redundancy
- Demonstrated that hidden states encode part-of-speech information in early-to-mid layers, with performance declining in later layers as representations become more task-specific
- Extended analysis to word sense disambiguation and textual entailment, showing that the model's internal mechanisms are task-driven rather than mirroring human reasoning
- Published at ICDMW 2024

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript
Frameworks & Tools: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas, MATLAB
Infrastructure: Git, Linux, CUDA, Docker, AWS, LaTeX
Methods: Hidden-state probing, Markov chain analysis, stochastic processes, spectral analysis, clustering (K-means, spectral, hierarchical), benchmark construction, statistical evaluation (bootstrap, permutation tests, cross-validation)